

Hoff Bayesian Statistics

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These are (fully reproducible!) R Markdown lecture notes for Peter D. Hoff, “A First Course in Bayesian Statistical Methods”¹, completed as part of a 1-semester independent study course. Only Chapters 1-8 are complete right now.

Each note includes summaries of chapter sections, with math and explanations modified for clarity and the occasional link to external resources. Many figures from the book are reproduced in a ggplot/tidyverse style, and some exercises at the end of each chapter are tackled (correctness not guaranteed).

If you find an error or would like to improve the notes, please submit a PR or open an issue!

1 Chapters

- Chapter 1: Introduction and examples²
- Chapter 2: Belief, probability, and exchangeability³
- Chapter 3: One-parameter models⁴
- Chapter 4: Monte Carlo approximation⁵
- Chapter 5: The Normal Model⁶
- Chapter 6: Posterior approximation with the Gibbs sampler⁷
- Chapter 7: The multivariate normal model⁸
- Chapter 8: Group comparisons and hierarchical modeling⁹
- Chapter 9: Linear regression¹⁰
- Chapter 10: Nonconjugate priors and Metropolis-Hastings algorithms¹¹

2 Final Project

As a small final project, R code for the basic binary relation version of the Infinite Relational Model was implemented, as described in Kemp et al. (2006), “Learning Systems of Concepts with an Infinite Relational Model”¹².

- Infinite Relational Model¹³

¹<https://peterhoff.io/book/>

²[1.qmd](#)

³[2.qmd](#)

⁴[3.qmd](#)

⁵[4.qmd](#)

⁶[5.qmd](#)

⁷[6.qmd](#)

⁸[7.qmd](#)

⁹[8.qmd](#)

¹⁰[9.qmd](#)

¹¹[10.qmd](#)

¹²<http://web.mit.edu/cocosci/Papers/Kemp-et-al-AAAI06.pdf>

¹³[irm.qmd](#)